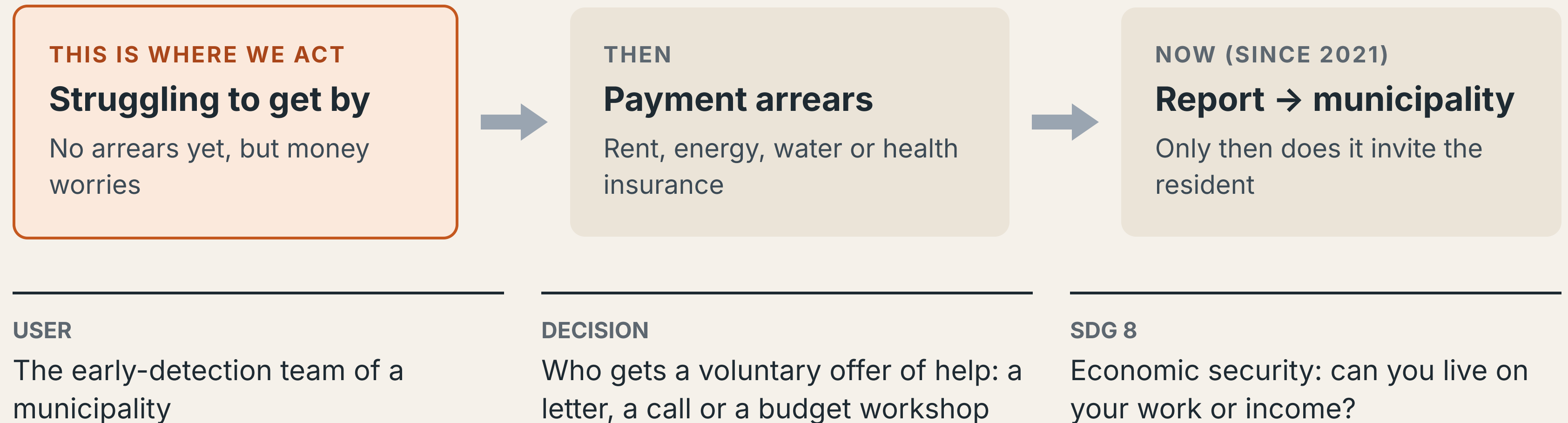


MODEL SHOWDOWN · HACKATHON 5 · SDG 8

Who struggles to get by?

Early detection for a Dutch municipality's debt-support team, with data from the European Social Survey (Netherlands, 2008–2023)

The signal only comes once the debt exists



1 in 10 struggles, so no accuracy

10%

1,407 of the 13,740 respondents find it (very) difficult to live on their household income.

A model that always says **"no"** is 90% accurate and finds **nobody**.

That is why we use F2

- Recall counts twice as much as precision.
- Missing someone who needs help is worse than one letter too many.
- Not recall alone: then flagging everyone would win.

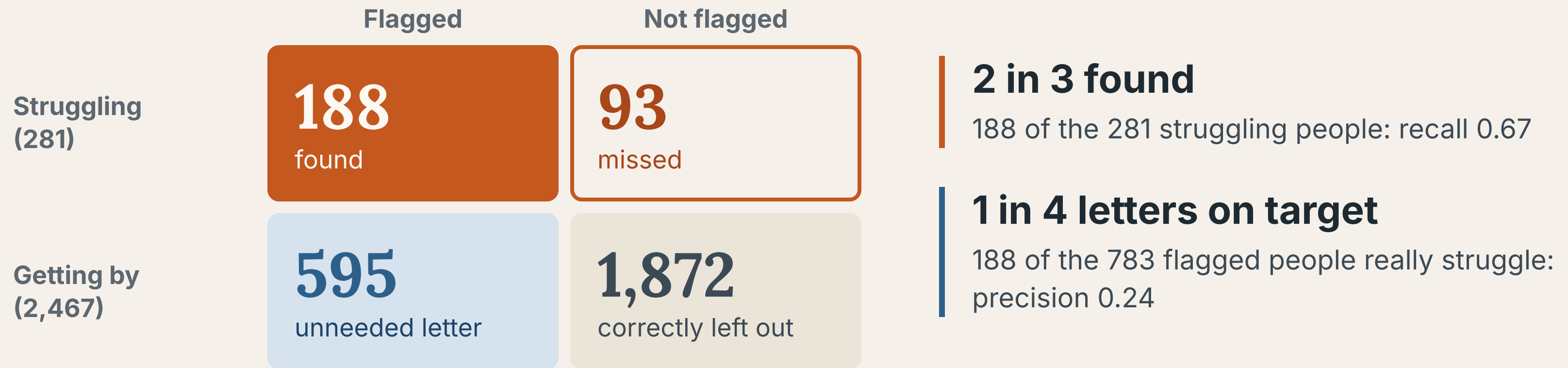
One fixed threshold for all models: flag someone when the chance is above 10%, the average.

Five candidates, same folds, same threshold

Model	CV F2	Test F2	Recall	Precision	Flagged
Always "no"	0.000 ± 0.000	0.00	0.00	0.00	0%
Simple rule	0.442 ± 0.014	0.45	0.48	0.35	14%
KNN (k = 301)	0.492 ± 0.017	0.46	0.53	0.31	18%
Logistic regression (C = 10)	0.510 ± 0.013	0.49	0.67	0.24	28%
Decision tree (depth 6)	0.465 ± 0.017	0.50	0.59	0.31	19%

Logistic regression finds **2 in 3** struggling people, against half for the simple rule. The decision tree scores slightly higher on the test set, but **clearly lower** in cross-validation (0.47 vs 0.51).

Logistic regression on the test set



Test set, used once · threshold: chance $\geq$ 10% (the average share of struggling people)

Our advice: logistic regression



Misses the fewest

93 of 281 missed; the decision tree misses 22 more. Best CV score: 0.51.



No overfitting

Train 0.52 · CV 0.51 · test 0.49.
The decision tree drops from 0.51 on train to 0.47 in CV.



Explainable

One weight per answer.
Unemployment benefit raises the chance; studying or a master's degree lowers it.



Only for a voluntary offer of help, next to the existing payment-arrears signals. Never for a decision with consequences for the resident.

Live: one resident through the model

MADE-UP RESIDENT

Feature	Value
Age	45
Household	3 people
Main activity	unemployed, looking
Main income	social benefits
Education	lower secondary
Lives in	a big city
Ever unemployed > 3 months	yes

83%

chance of struggling to get by. Above the 10% threshold, so: **send an offer of help.**

Then change it live: 72 years old, retired, lives alone, pension → **4%**, no offer.

Who does the model miss?

27%

of struggling **retired people** are found. On paper they look like people who are fine.

46%

of struggling **working people** are found. Rent and debts are not in the data.

77%

Born outside NL: flagged more often, but also struggle 3× as often. Found better than people born in NL (64%).

≈

Men and women are found about equally well: recall 64% and 68%.

Country of birth is still **partly hidden** in benefits and education: our features predict “born outside NL” with ROC-AUC 0.68 (0.5 = no information).

Recall (share found) per group on the test set · sex and country of birth are not model inputs, only used for this check

Does the model still work today?

2008–2018

61–79%

of struggling people found. 9–15% struggle.

2020 AND 2023

±45%

of struggling people found. Only 4–6% struggle; 1 in 9 letters is on target.

The model knows the past better than the present. **Test it on recent municipal data first**, then use it.

THE DECISION WE DEFEND HARDEST

Income, sex and country of birth stay out.

Income

A municipality does not know its residents' income. 13% of respondents did not give it either.

Sex and origin

Selecting on sex or origin is not allowed. The Dutch childcare-benefits scandal showed where that leads.

The price

With income: F2 **0.59**. Without: **0.51**.
Eight points, more than the gap between our models.

Better a slightly weaker model that only uses what the municipality may and can know.

How to use it, and how not to

Do

- A voluntary letter, call or budget workshop
- Next to the payment-arrears signals
- Test on recent municipal data first, with a DPIA

Don't

- Fraud checks, sanctions, benefits or credit
- As the only way to find people
- For retired people without extra attention

Questions?